

Exploring Hallucination-Aware Extensions of HIRE for LVLMs

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Abstract

Base on hire, we try vti and interpretability

1. Introduction

Multimodal Large Language Models (LVLMs) have achieved remarkable progress in image understanding by leveraging the world knowledge of Large Language Models (LLMs). As a result, LVLMs can be applied to a wider range of vision-language tasks with stronger reasoning capabilities.

However, LVLMs still suffer from hallucinations, generating responses that describe non-existent objects or incorrectly interpret visual content. Such hallucinations reduce model reliability, motivating extensive research on hallucination mitigation.

Recent studies assume that hallucination-related information can be distinguished in hidden representation spaces and have achieved promising results through contrastive learning and representation editing. However, these approaches typically require additional training and often construct hallucination data based on the observation that hallucinations occur more frequently under degraded image quality.

In this project, we adopt a training-free hallucination mitigation framework and further introduce an additional analysis module to investigate hallucination-related representations.

2. Related Work

2.1. HIRE

Recent studies have shown that hallucination-related information can be identified in the hidden representations of LVLMs. Based on this observation, HIRE [4] proposes a representation-level hallucination manipulation framework consisting of an editor and a router.

HIRE learns to disentangle semantic and hallucination-related representations through image perturbation. Clean and perturbed images are encoded into latent represen-

tations, and a clustering objective is used to separate hallucination-related and reliable latent representations while preserving semantic information. Based on the learned representation space, HIRE derives editing directions that can either suppress or amplify hallucination behavior through a controllable editing scale.

To improve efficiency, HIRE further introduces a router that selectively applies editing to a subset of tokens. The router is trained using Dynamic Preference Optimization with randomly generated editing trajectories, enabling efficient hallucination mitigation while reducing computational cost.

Although HIRE achieves strong hallucination mitigation performance, it requires additional training for representation editing and routing. Furthermore, the relationship between perturbation-induced representations and hallucination formation remains insufficiently explained.

2.2. VTI

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2.3. Interpretable Model Structure

Interpretable neural network architectures have been widely studied to improve model transparency. One representative approach is the Concept Bottleneck Model (CBM) [2], which introduces an intermediate concept layer between hidden representations and final predictions. By explicitly modeling concept activations, CBMs enable interpretation of how information contributes to model decisions.

However, CBMs require concept-level supervision, which is unavailable for hallucination-related representations in LVLM hidden spaces. To address this limitation, we adopt prototype learning methods [1, 3]. Prototype learning constructs representative prototypes for each class and learns discriminative feature patterns directly from hidden representations. As a result, meaningful concepts can be extracted without explicit representation-level annotations.

In this work, we use prototype learning to construct representation-level hallucination concepts and train a CBM on the resulting prototype activations. This enables concept-level analysis of hallucination-related representa-

tions.

3. Method

Here we talk about what we do.

3.1. Sementic Representation in Training free

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3.1.1. Problem Setup

use vti

3.1.2. Prompt Filtering via Subspace Projection

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3.2. Representation-level Hallucination Interpretability

Existing hallucination mitigation methods focus on reducing hallucinated outputs but provide limited interpretability of hallucination-related representations. To address this limitation, we introduce a prototype-based CBM that learns hallucination concepts in hidden representation space and analyzes representations through concept activations.

3.2.1. Prototype-based Concept Learning

CBMs require concept-level supervision, but hallucination concepts are not explicitly annotated in hidden representation space. To obtain interpretable concepts, we learn layer-wise prototypes corresponding to Safe and Existence hallucination categories and construct a prototype bank across all transformer layers.

Given a hidden representation h_i^l from transformer layer l , a prototype encoder projects it into a lower-dimensional embedding space z_i^l . The encoded representation is compared with learnable concept prototypes using cosine similarity $s_{i,l,c}$, where $c \in \{\text{safe}, \text{existence}\}$ denotes the concept category.

To obtain discriminative concept representations, we jointly optimize classification, clustering, and separation objectives. The classification loss encourages correct concept assignment, the cluster loss pulls samples toward the corresponding prototype, and the separation loss pushes samples away from other concept prototypes.

Algorithm 1 summarizes the prototype learning procedure.

3.2.2. Concept Bottleneck Module

After learning concept prototypes, we train a CBM. Instead of directly predicting hallucination labels from hidden representations, the CBM estimates concept activations from prototype similarities, providing an interpretable bottleneck between representations and hallucination prediction.

Algorithm 1 Layer-wise Prototype Learning for Safe / Existence Concepts

Require: Layer-wise hidden representations $\mathcal{H} = \{(H_i, y_i)\}_{i=1}^N$, where $y_i \in \{\text{safe}, \text{existence}\}$

Require: $H_i = [h_i^1, h_i^2, \dots, h_i^L]$, where $h_i^l \in \mathbb{R}^{d_h}$

Require: Learnable layer-wise prototypes $\{p_{l,c}\}_{l=1, c \in \{\text{safe}, \text{exist}\}}^L$

Require: Prototype encoder $f_\theta : \mathbb{R}^{d_h} \rightarrow \mathbb{R}^{d_z}$

```

1: for each training epoch do
2:   for each mini-batch  $\mathcal{B} \subset \mathcal{H}$  do
3:     for each sample  $(H_i, y_i) \in \mathcal{B}$  do
4:       for each layer  $l = 1, \dots, L$  do
5:         Encode layer representation

```

$$z_i^l \leftarrow f_\theta(h_i^l)$$

```

6:         Normalize representation and proto-
           types

```

$$\tilde{z}_i^l \leftarrow \frac{z_i^l}{\|z_i^l\|_2}, \quad \tilde{p}_{l,c} \leftarrow \frac{p_{l,c}}{\|p_{l,c}\|_2}$$

```

7:         Compute prototype similarity

```

$$s_{i,l,c} = (\tilde{z}_i^l)^\top \tilde{p}_{l,c}$$

```

8:       end for
9:       Aggregate layer-wise similarities

```

$$\ell_{i,c} = \frac{1}{L} \sum_{l=1}^L s_{i,l,c}$$

```

10:      Compute prototype loss

```

$$\mathcal{L}_i = \mathcal{L}_{cls}^i + \lambda_c \mathcal{L}_{cluster}^i + \lambda_s \mathcal{L}_{sep}^i$$

```

11:    end for
12:    Compute mini-batch loss

```

$$\mathcal{L}_{proto} = \frac{1}{|\mathcal{B}|} \sum_{(H_i, y_i) \in \mathcal{B}} \mathcal{L}_i$$

```

13:    Update encoder and prototypes
14:  end for
15: end for
16: return  $f_\theta, \{p_{l,c}\}$ 

```

For each sample, prototype similarities from all transformer layers are aggregated into a concept vector v_i . The vector is processed by an MLP g_ϕ , and concept activations $\hat{c}_i$ are obtained through a sigmoid function.

To improve concept interpretability, we optimize the CBM using both prediction and concept supervision objectives. Binary cross-entropy encourages correct concept prediction, while sparsity regularization discourages multiple

Algorithm 2 Layer-wise Concept Bottleneck Training with Prototype Activations

Require: Layer-wise hidden representations $\mathcal{H}$

Require: Trained prototype bank $\{p_{l,c}\}$

Require: Prototype encoder f_θ

Require: Concept bottleneck layer g_ϕ

```

1: for each training epoch do
2:   for each mini-batch do
3:     for each sample do
4:       for each layer  $l$  do
5:         Encode representation

```

$$z_i^l = f_\theta(h_i^l)$$

```

6:       Compute similarity

```

$$s_{i,l,c} = \cos(z_i^l, p_{l,c})$$

```

7:     end for
8:   Aggregate concept vector

```

$$v_i = \left[\frac{1}{L} \sum_l s_{i,l,1}, \dots, \frac{1}{L} \sum_l s_{i,l,K} \right]$$

```

9:   Predict concept activations

```

$$\hat{c}_i = \sigma(g_\phi(v_i))$$

```

10:   Compute CBM loss
11: end for
12:   Update CBM parameters
13: end for
14: end for
15: return  $g_\phi$ 

```

concepts from being activated simultaneously.

$$\mathcal{L}_{CBM} = \mathcal{L}_{CE} + \lambda_{BCE} \mathcal{L}_{BCE} + \lambda_{sparse} \mathcal{L}_{sparse}$$

Algorithm 2 summarizes the CBM training procedure. The learned concept activations provide an interpretable view of hidden representations and enable hallucination analysis at the representation level. By preserving layer-wise prototype similarities, the proposed framework allows investigation of how hallucination-related information evolves.

4. Experiments

We test our project

Table 1. Hallucination evaluation on the COCO validation subset (100 images). Lower CHAIRs/CHAIRi is better, while higher Recall is better.

Model	CHAIRs ↓	CHAIRi ↓	Recall ↑
LLaVA	0.5200	0.1409	0.7940
HIRE	0.5000	0.1275	0.7209
LLaVA+CBM	0.5200	0.1417	0.7859
HIRE+CBM	0.5000	0.1276	0.7127

Table 2. Correlation between CBM Existence Metrics and CHAIR Scores

Metric	CHAIRi	CHAIRs
Avg. Existence Score	0.274	0.279
Max. Existence Score	0.291	0.330
Avg. Prototype Similarity	0.273	0.280
Max. Prototype Similarity	0.290	0.319

4.1. Hallucination Interpretability

4.1.1. Experimental Setup

We evaluate the CBM on the generative subset of the AMBER dataset using LLaVA-1.5-7B and HIRE.

The dataset[5] contains 9,848 Existence hallucination samples and 5,599 Safe samples. Hidden representations are extracted from all 32 transformer layers, and prototype similarity is measured using cosine similarity.

4.1.2. CBM Validation

Minimal Impact on Generation Table 1 reports CHAIR and Recall scores on the COCO validation subset. As expected, introducing the CBM module causes negligible changes in generation performance. The differences between the original models and their CBM counterparts remain minimal across all metrics. This is expected because the proposed framework is designed for representation analysis rather than hallucination mitigation.

Correlation with Hallucination Metrics To investigate whether the learned concepts are related to hallucination behavior, we compute correlations between CBM concept activations, prototype similarities, and CHAIR metrics. Table 2 presents the correlation coefficients. Although the correlations are moderate, all metrics exhibit positive relationships with both CHAIRi and CHAIRs. These observations suggest that the learned concepts capture hallucination-related characteristics in representation space.

4.1.3. Representation Analysis

Consistent Concept Separation To investigate whether the learned concepts are consistently separated, we analyze prototype margins and cross-layer prototype similarities.

As shown in Figure 1, Safe and Existence concepts begin to separate after approximately the third transformer layer,

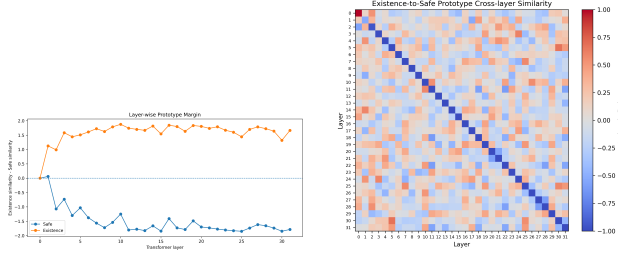


Figure 1. Concept separation analysis. Left: layer-wise prototype margin between Safe and Existence concepts. Right: cross-layer prototype similarity matrix.

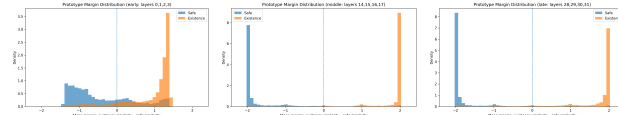


Figure 2. Prototype activation distributions across transformer layers.

and the margin converges around the eighth layer. This indicates that the learned prototypes capture distinct representation patterns associated with hallucination behavior.

The cross-layer similarity matrix further shows strong diagonal patterns, suggesting that each layer learns a stable and distinct concept representation. Together, these results indicate consistent concept separation throughout the network.

Stronger Signals in Deeper Layers We further analyze prototype activation distributions across early, middle, and late transformer layers.

Figure 2 shows substantial overlap between Safe and Existence concepts in early layers. However, the separation becomes increasingly clear in middle and deeper layers.

This trend suggests that hallucination-related information becomes progressively more distinguishable as representations move toward higher semantic abstraction levels.

4.1.4. Limitation

Although the proposed framework provides meaningful concept-level interpretations of hallucination-related representations, several limitations remain.

As shown in Figure 3, concept activations exhibit only moderate correlations with hallucination metrics. Furthermore, although Existence activations become increasingly pronounced in deeper layers, the learned concepts are not directly aligned with HIRE’s editing objective.

We believe this limitation arises because our framework focuses exclusively on Existence hallucinations, whereas HIRE addresses multiple hallucination categories simultaneously. Consequently, concept activations capture hallucination-related characteristics but remain weakly connected to actual hallucination mitigation performance.

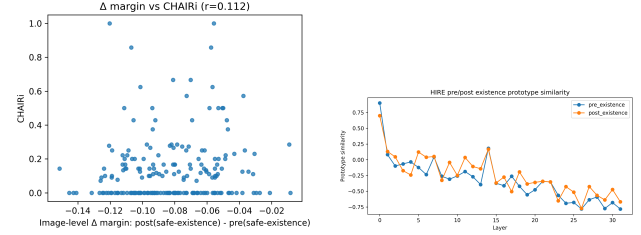


Figure 3. Limitations of the proposed framework. Left: relationship between concept activation changes and CHAIRi scores. Right: layer-wise Existence concept activations.

Table 3. Inference overhead introduced by the layer-wise CBM module.

Model	Runtime (s)	With CBM (s)	Overhead
LLaVA	60	352	5.87×
HIRE	139	1212	8.72×

The proposed framework also introduces additional computational overhead due to layer-wise representation extraction and prototype similarity computation.

As shown in Table 3, the additional analysis module increases inference time by approximately 5.9× to 8.7×.

Future work will focus on extending the concept space beyond Existence hallucinations and integrating concept information directly into representation editing frameworks.

5. Conclusion

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References

- [1] Sangwon Kim, Jaeyeal Nam, and Byoung Chul Ko. Vit-net: Interpretable vision transformers with neural tree decoder. In *International conference on machine learning*, pages 11162–11172. PMLR, 2022. 1
- [2] Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. Concept bottleneck models. In *International conference on machine learning*, pages 5338–5348. PMLR, 2020. 1
- [3] Gurmail Singh and Kin-Choong Yow. These do not look like those: An interpretable deep learning model for image recognition. *IEEE Access*, 9:41482–41493, 2021. 1
- [4] Wei Suo, Hanzu Zhang, Lijun Zhang, Ji Ma, Peng Wang, and Yanning Zhang. Hallucination-aware intermediate representation edit in large vision-language models. *arXiv preprint arXiv:2603.29405*, 2026. 1
- [5] Junyang Wang, Yuhang Wang, Guohai Xu, Jing Zhang, Yukai Gu, Haitao Jia, Jiaqi Wang, Haiyang Xu, Ming Yan, Ji Zhang, et al. Amber: An llm-free multi-dimensional benchmark for mllms hallucination evaluation. *arXiv preprint arXiv:2311.07397*, 2023. 3